

Econometrics for Finance

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This course offers an introductory overview of econometric methods and concepts specifically relevant to the field of finance. My goal is to give you a broad understanding of these techniques and their implementation in Python, but please note that this is not an exhaustive course. For more comprehensive coverage and practical applications, you can consider further specialized courses or books.

1 Course Format

The course consists of four in-person lectures in which I cover the fundamental concepts and methods. In class, we also discuss how to implement these techniques in Python: we code together and work through problems, so bring a laptop.

There are no graded assignments and no exam. The course is there to give you a solid foundation in econometric methods and their relevance in finance, and to support the thesis you write afterwards. What you get out of it depends on how much you engage with the material during the lectures and the research phase. If you have any questions or require further information, please feel free to reach out to me.

2 Prerequisites

Officially, there are no prerequisites for this course. However, it is highly recommended that you have a basic understanding of probability and statistics, which will be beneficial for your learning experience. I do my best to explain the concepts in a clear and accessible manner, but having some prior knowledge in these areas will help you grasp the material more effectively.

Additionally, while this course focuses on econometric methods and their implementation in Python, it is not a programming course. Therefore, having some familiarity with Python will be advantageous. If you are not familiar with Python, I strongly recommend that you look at the materials I post on Canvas. These materials will help you get started and give you the skills you need to implement the econometric methods we cover in the course.

Software setup

Please arrive at the first session with a working Python environment. We do not spend class time on installation.

- **Python 3.10 or newer.** The simplest route is the [Anaconda](#) distribution, which bundles almost everything below.
- **Packages:** `numpy`, `pandas`, `statsmodels`, `linearmodels`, `matplotlib`. With Anaconda, only `linearmodels` typically needs to be installed separately (`pip install linearmodels`).
- **An editor:** Jupyter Notebook, JupyterLab, or VS Code. Any of these is fine; use whichever you are most comfortable with.
- **No-install alternative:** [Google Colab](#) runs in the browser with the packages above already available. This is a reasonable fallback if you run into installation problems, though for thesis work a local setup is more convenient.

To check that you are ready, run `import numpy, pandas, statsmodels, linearmodels` without errors before the first lecture. If it fails, contact me in advance rather than on the day.

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3 Course Schedule

The course schedule is shown in Table 1. Rooms and the exact timeline will be posted on Canvas.

Table 1: Course Schedule

Session	Content	Class time	Date
1	Introduction & Cross-section I	13:15 - 15:00	Thu 3 Sep
2	Cross-section II & IV	15:15 - 17:00	Tue 8 Sep
3	Panel Methods & Diff-in-Diff I	13:15 - 15:00	Tue 15 Sep
4	Diff-in-Diff II & RD & Portfolio Sorts & Event Study	13:15 - 16:00	Tue 22 Sep

Note: Rooms and the exact timeline will be posted on Canvas.

Note that Session 4 covers four topics in under three hours. Portfolio sorts and event studies are treated at the level of the basic recipe plus references, which should be enough to get you started, but you will likely need to read further if your thesis leans heavily on either method.

4 Research Phase

After completing the course, you will have the opportunity to work on your own project, and I will be available to provide support during this period. You can book a time slot through this [link](#). You can also use this time to ask any questions you may have about the course material.

Please note that while I can offer guidance and support in the areas of econometrics and coding, I may not know all fields of finance. However, I will do my best to help you with the econometric and technical aspects of your project.

I am glad to work with you on your project, and I look forward to seeing what you find.

P.S.: Chapter 15 of [Brooks \(2019\)](#) provides a good insight into the research process.

5 General Information

- In your actual research or work, you will face many problems that you have not seen before. Therefore, you should learn how to find the answer to your questions. Part of what I am trying to do here is help you build that skill. Keep in mind that there is no unique process for answering your questions.
- Before asking a question:
 - Ask [Google](#)
 - Use any AI tools that you like!
 - Search in [stackoverflow](#)
- On AI tools: you can never justify a choice in your research with “I asked an AI and it said that...”. The argument has to stand on its own. AI models are powerful and genuinely useful, but they cannot replace your own thinking: read their answers carefully, check them against a source you trust, and make sure you can explain every step of your analysis yourself.
- Suggested textbooks:
 - *Introductory Econometrics for Finance* ([Brooks 2019](#))
 - * There is a free [Python Guide](#) and [R Guide](#) for this book.
 - *Mostly Harmless Econometrics: An Empiricist’s Companion* ([Angrist and Pischke 2009](#))
 - *Introductory Econometrics: A Modern Approach* ([Wooldridge 2015](#))
- Suggested courses:
 - A full course in econometrics [[part 1](#), [part 2](#)].

- Data sources:
 - [SHoF Data Center](#)
 - You can access the databases distributed via [WRDS](#), LSEG Workspace (formerly Refinitiv Eikon) and [S&P Capital IQ](#) through the [Library](#)
 - You can access the SDC Platinum through the SHoF data room.
 - You can also use some other free data sources.
 - Sometimes it is possible to crawl and download data from free websites. The quality of such data varies, so budget time for thorough checking and cleaning before you use it. Treat this as a fallback rather than the plan your thesis depends on.

References

- Angrist, J.D., and J.S. Pischke.** 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Brooks, Chris.** 2019. *Introductory Econometrics for Finance*. . 4 ed., Cambridge University Press.
- Wichmann, Robert, and Chris Brooks.** 2019. *R Guide to Accompany Introductory Econometrics for Finance*. SSRN.
- Wooldridge, J.M.** 2015. *Introductory Econometrics: A Modern Approach*. Cengage Learning.